



EntitySpaces 2027 – Quick Start Guide

Version for developers who want to get started in less than 15 minutes.

1. Requirements

- **.NET Framework 4.8** or **.NET 8/9/10** (depending on your project).
- **Visual Studio 2022** (or 2019) with Desktop/Web development workloads installed.
- **EntitySpaces Studio 2024** (standalone code generator).

2. Download and Install EntitySpaces Studio

1. Go to the GitHub repository:
<https://github.com/paulcordova/EntitySpaces>
and download the **EntitySpacesStudio_2024.8.30.0.zip** file (or the latest version available).
2. Extract the contents to a local folder (e.g., **C:\EntitySpaces\Studio**).
3. Run **StandAlone.exe** as Administrator (Windows may prompt for permissions).

✓ **Tip:** The Studio is a WinForms application that runs independently of Visual Studio. You can use it on any machine.

3. Connect to your Database

Once the Studio is open:

1. Go to the **Settings** → **Connection** tab.
 2. Select the **Driver** corresponding to your database engine (SQL Server, PostgreSQL, MySQL, SQLite, Oracle, Firebird).
 3. Click **Configure Connection** to build the connection string, or enter it manually.
 4. Test the connection using the **Test Connection** button.
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Example connection strings:

Database	Example Connection String
SQL Server	User ID=sa;Password=...;Initial Catalog=Northwind;Data Source=localhost;TrustServerCertificate=True;
PostgreSQL	Host=localhost;Port=5432;Database=Northwind;Username=postgres;Password=...;SSL Mode=Require;
MySQL	Server=localhost;Port=3306;Database=Northwind;Uid=root;Pwd=...;SslMode=Required;
SQLite	Data Source=C:\Northwind.db3;Version=3;Foreign Keys=True;
Oracle	Data Source=...;User Id=...;Password=...; (additional wallet configuration required for ATP)

4. Generate C# or VB.NET Classes

1. Go to the **Templates** tab.
2. Select your preferred language (C# or VB.NET) and the "**Generated Classes Master**" template.
3. In the dialog that appears:
 - Define the **Namespace** (e.g., **BusinessObjects**).
 - Choose the tables/views you want to generate.
 - In the **Advanced** tab, review options such as **Generate a Single File** (recommended).
4. Click **OK**. The Studio will generate the files in the folder you specified (by default, in the **Generated** and **Custom** subfolders).
5. Repeat the process with the "**Custom Classes Master**" template (this is generated only once; you will place your business logic here).

5. Add NuGet Packages to your Project

Open your project in Visual Studio and add the provider package you need.

From the Package Manager Console:

```
# Example for SQL Server
Install-Package EntitySpaces.ORM.SqlServer.NET

# For PostgreSQL
Install-Package EntitySpaces.ORM.PostgreSQL.NET

# For MySQL
Install-Package EntitySpaces.ORM.MySQL.NET

# For SQLite
Install-Package EntitySpaces.ORM.SQLite.NET

# For Oracle Managed Client
Install-Package EntitySpaces.ORM.OracleManagedClient.NET

# For Firebird
Install-Package EntitySpaces.ORM.Firebird.NET
```

These packages include all necessary dependencies (`EntitySpaces.Core`, `EntitySpaces.Interfaces`, `EntitySpaces.DynamicQuery`, the Loader, and the specific provider).

6. Configure the Loader and Connection in your Code

EntitySpaces requires you to register the **Loader** (Medium Trust or Standard) and the **connection string**.

At the start of your application (e.g., `Program.Main`, `Global.asax`, or the main form constructor):

```
using EntitySpaces.Interfaces;

// 1. Register the Loader (Medium Trust version is recommended)
esProviderFactory.Factory = new EntitySpaces.LoaderMT.esDataProviderFactory();

// 2. Configure the connection
esConnectionElement conn = new esConnectionElement();
conn.Provider = "EntitySpaces.SqlClientProvider"; // Change according to your
provider
conn.ConnectionString = "User ID=...;Password=...;Initial Catalog=Northwind;Data
Source=localhost;";
conn.SqlAccessType = esSqlAccessType.DynamicSQL; // or StoredProcedure
conn.DatabaseVersion = "2016"; // Optional, auto-detected if
omitted

// 3. Add the connection and set it as the default
esConfigSettings.ConnectionInfo.Connections.Add(conn);
esConfigSettings.ConnectionInfo.Default = "Northwind"; // any name you prefer
```

Alternative: Use an `app.config` / `web.config` file with the `<EntitySpaces>` section. The complete guide is available in the repository's README.

7. Write your First Query

With the generated classes and configuration in place, you can start loading data:

```
// Load all customers
CustomersCollection customers = new CustomersCollection();
if (customers.LoadAll())
{
    foreach (Customers cust in customers)
    {
        Console.WriteLine($"{cust.CustomerID} - {cust.CompanyName}");
    }
}

// Query with filter (Fluent API)
CustomersCollection coll = new CustomersQuery("c", out var c)
    .Where(c.Country == "USA")
    .OrderBy(c.CompanyName.Ascending)
    .ToCollection<CustomersCollection>();

// Basic CRUD
Customers newCust = new Customers();
newCust.CustomerID = "NEWCO";
newCust.CompanyName = "Nueva Empresa";
newCust.Save(); // INSERT

newCust.CompanyName = "Nueva Empresa S.A.";
newCust.Save(); // UPDATE

newCust.MarkAsDeleted();
newCust.Save(); // DELETE
```

8. Usage with Wisej.NET

Wisej.NET is a framework that allows you to build web applications using the same logic and event model as Windows Forms. EntitySpaces integrates seamlessly:

- Reuse all your generated classes and existing business logic.
- The connection and transaction patterns are identical to those of a desktop application.
- The ORM correctly handles connection pooling and concurrency in the web environment (thanks to the thread-safe parameter cache and transaction management in **finally** blocks).

Example in a Wisej control:

```
// In a Button Click event
private void buttonLoad_Click(object sender, EventArgs e)
{
    var products = new ProductsCollection();
    products.Query.Where(products.Query.UnitPrice > 10);
    products.Query.Load();

    dataGridView1.DataSource = products;
}
```

Refer to the [official Wisej.NET documentation](#) for more details on deployment.

9. Common Troubleshooting

Issue	Solution
"Cannot find EntitySpaces.Core assembly"	Ensure you have installed the correct NuGet package and that it is referenced in your project.
"Error connecting to SQLite"	Verify that the connection string includes <code>Version=3;Foreign Keys=True;</code> and that the .db3 file exists.
Studio doesn't show tables	In the Studio, make sure you have selected the correct Catalog/Database from the dropdown after connecting.
Architecture error when running Studio	The Studio is compiled for x86 ; do not run it in AnyCPU mode. If using native dependencies (SQLite, Oracle), ensure you have the x86 version.
Concurrency exception on save	EntitySpaces throws <code>esConcurrencyException</code> for key violations, deadlocks, etc. Catch it and handle the conflict according to your business logic.

10. Additional Resources

- **GitHub Repository:** <https://github.com/paulcordova/EntitySpaces>
 - Source code, complete examples, change history.
- **Extended Documentation:** Inside the repository, you will find the `README.md` with all technical details, including metadata migration, providers, and best practices.
- **Forum / Issues:** Use GitHub Issues to report problems or suggest improvements.
- **Consulting:** Contact paul.cordova@netstep.cl for professional support or training.

That's it! With these steps, your environment is set up and you are ready to start developing with EntitySpaces 2027.

Last updated: July 2026 – Compatible with .NET 10.